CSE4203: Computer Graphics

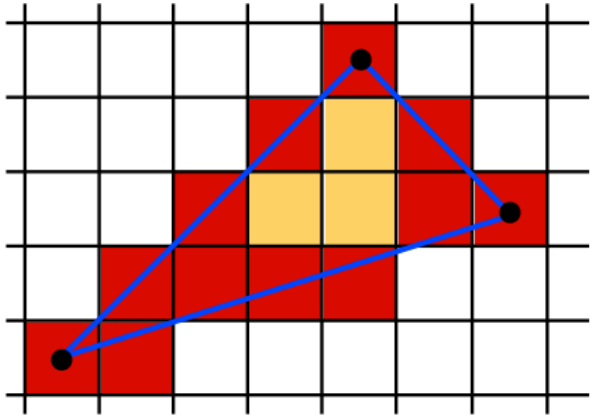
Lecture – 6 (part - C)

Graphics Pipeline

Outline

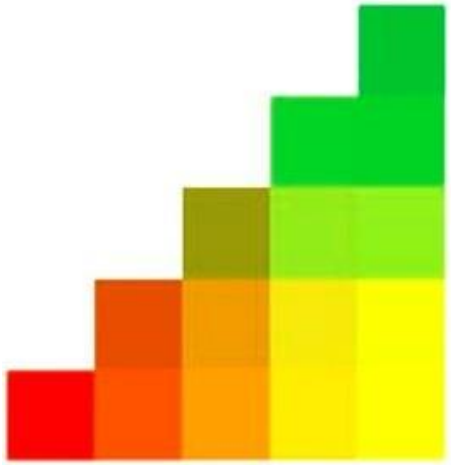
- Barycentric Interpolation
- Rasterizing a triangle

Triangle Rasterization

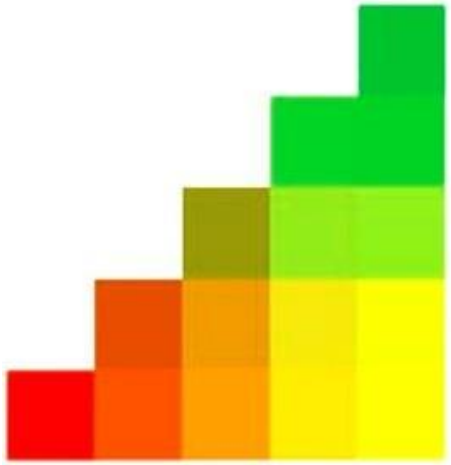


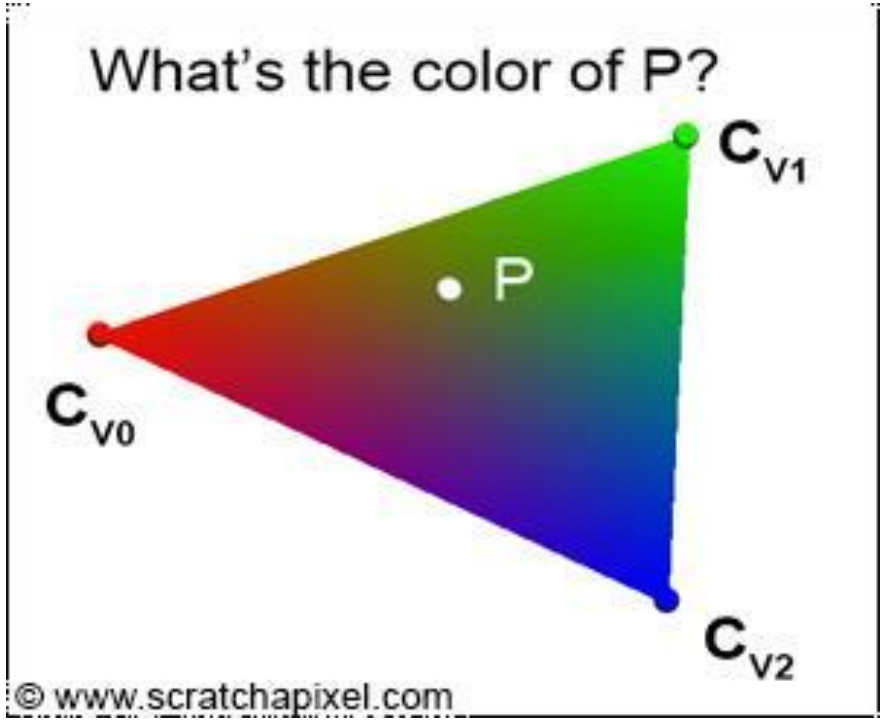
Use Midpoint Algorithm for edges and fill in?

Triangle Rasterization

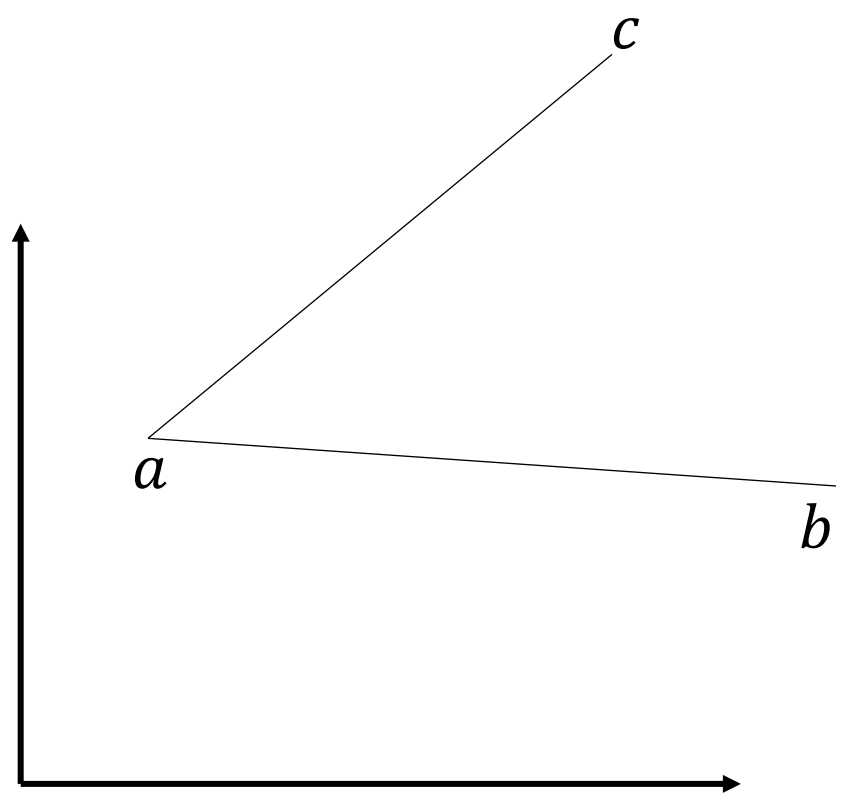


Triangle Rasterization





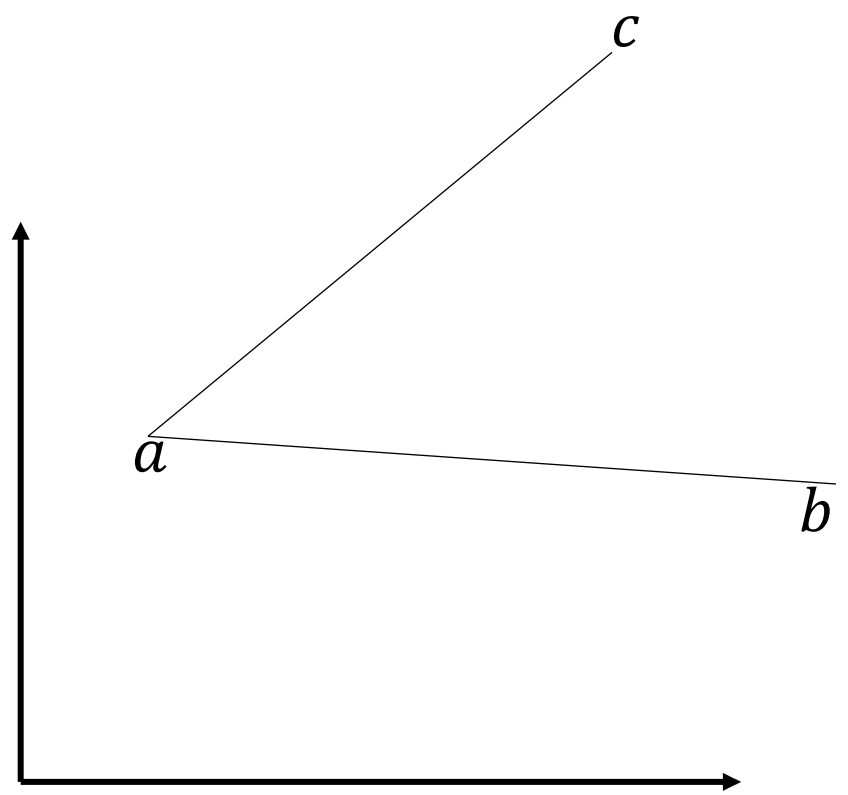
Barycentric Coordinate



$$P(x, y) = \boldsymbol{O} + x\boldsymbol{X} + y\boldsymbol{Y}$$

$$P(\beta, \gamma) = a + \beta\ ? + \gamma\ ?$$

Barycentric Coordinate



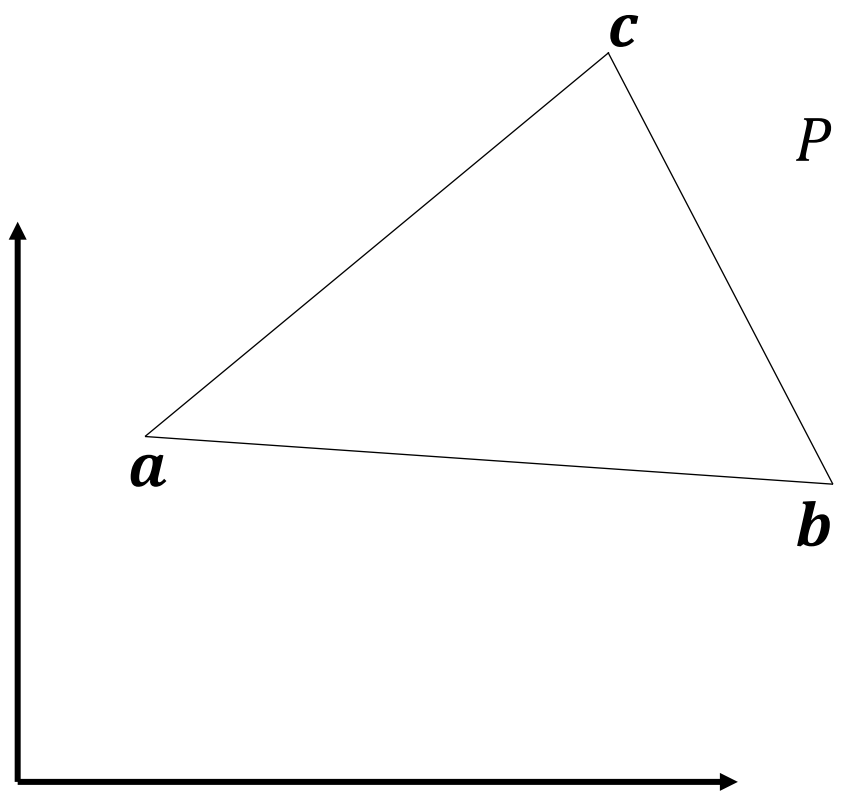
$$P(x, y) = \boldsymbol{O} + x\boldsymbol{X} + y\boldsymbol{Y}$$

$$P(\beta, \gamma) = a + \beta\ ? + \gamma\ ?$$

$$= a + \beta\ (b - a) + \gamma\ (c - a)$$

$$= a + \beta b - \beta a + \gamma c - \gamma a$$

Barycentric Coordinate



$$P(x, y) = \boldsymbol{O} + x\boldsymbol{X} + y\boldsymbol{Y}$$

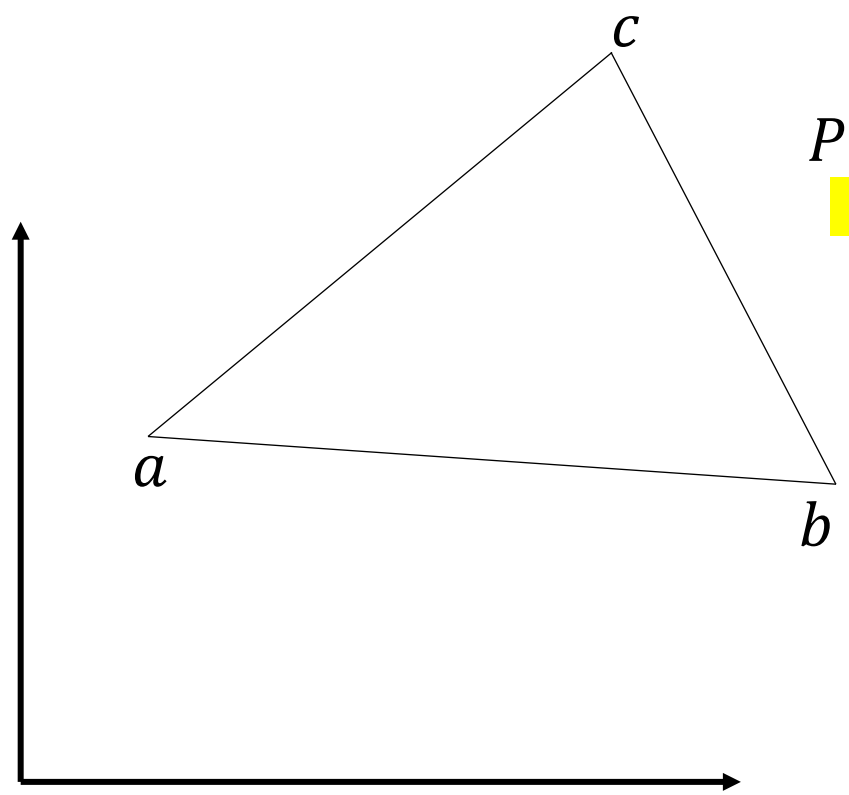
$$P(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}) = a + \beta\ ? + \gamma\ ?$$

$$= a + \beta\,(b - a) + \gamma\,(c - a)$$
$$= a + \beta b - \beta a + \gamma c - \gamma a$$
$$= a + \beta b - \beta a + \gamma c - \gamma a$$
$$= a(1 - \beta - \gamma) + b\beta + c\gamma$$
$$= \alpha\boldsymbol{a} + \beta\boldsymbol{b} + \gamma\boldsymbol{c}$$

$$\alpha = (1 - \beta - \gamma)$$
$$\Rightarrow \alpha + \beta + \gamma = 1$$

In a barycentric coordinate system: *location of a point is specified by reference to a triangle for points in a plane.*

Barycentric Coordinate



$$P(x, y) = \boldsymbol{O} + x\boldsymbol{X} + y\boldsymbol{Y}$$

$$P(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}) = \alpha\boldsymbol{a} + \beta\boldsymbol{b} + \gamma\boldsymbol{c}$$

Barycentric

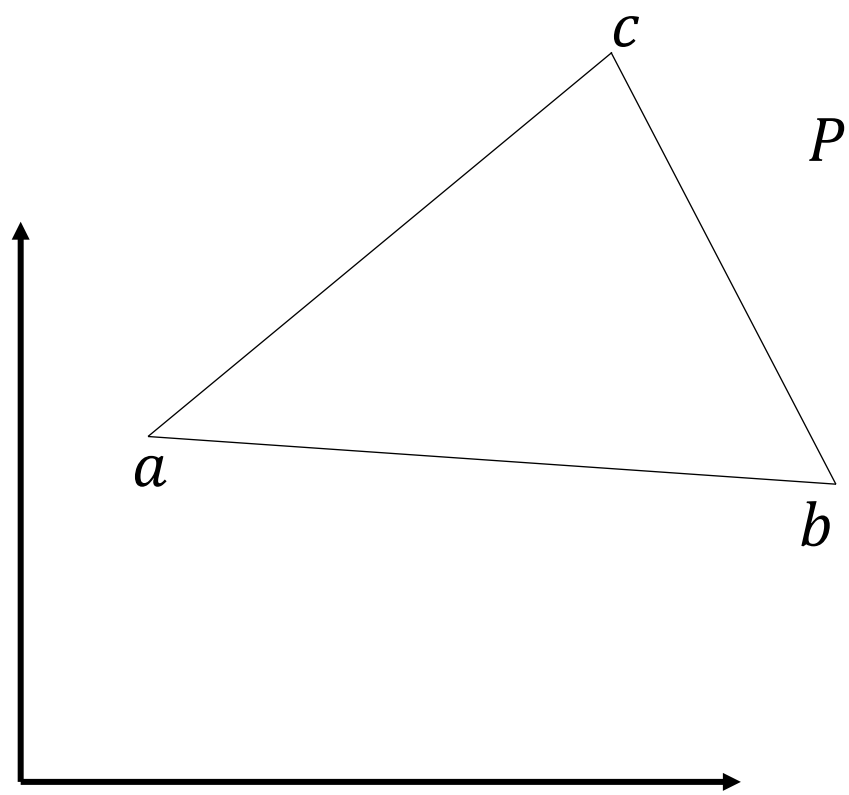
$$\alpha + \beta + \gamma = 1$$

$$0 < \alpha < 1,$$

$$0 < \beta < 1,$$

$$0 < \gamma < 1.$$

Barycentric Coordinate



$$P(x, y) = \boldsymbol{O} + x\boldsymbol{X} + y\boldsymbol{Y}$$

$$P(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}) = \alpha\boldsymbol{a} + \beta\boldsymbol{b} + \gamma\boldsymbol{c}$$

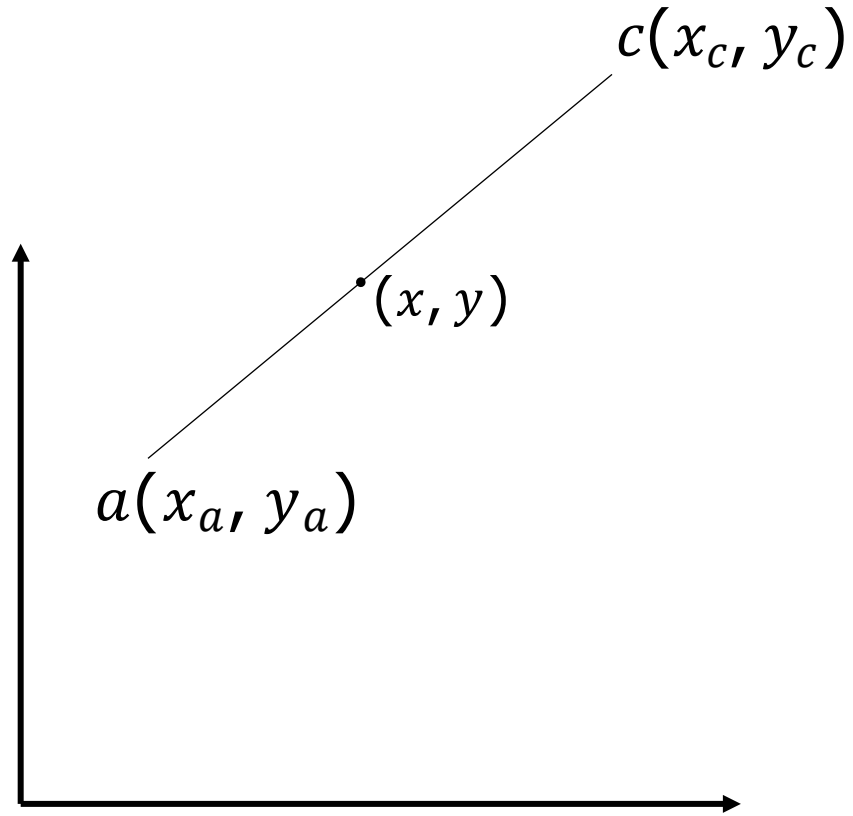
$$\alpha + \beta + \gamma = 1$$

$$0 < \alpha < 1,$$
$$0 < \beta < 1,$$
$$0 < \gamma < 1.$$

Cartesian $\rightarrow$ *Barycentric*

$$P(x, y) \rightarrow P(\alpha, \beta, \gamma)$$

Cartesian → Barycentric



$$y = m\,x + b$$

$$\Rightarrow y - m\,x - b = 0$$

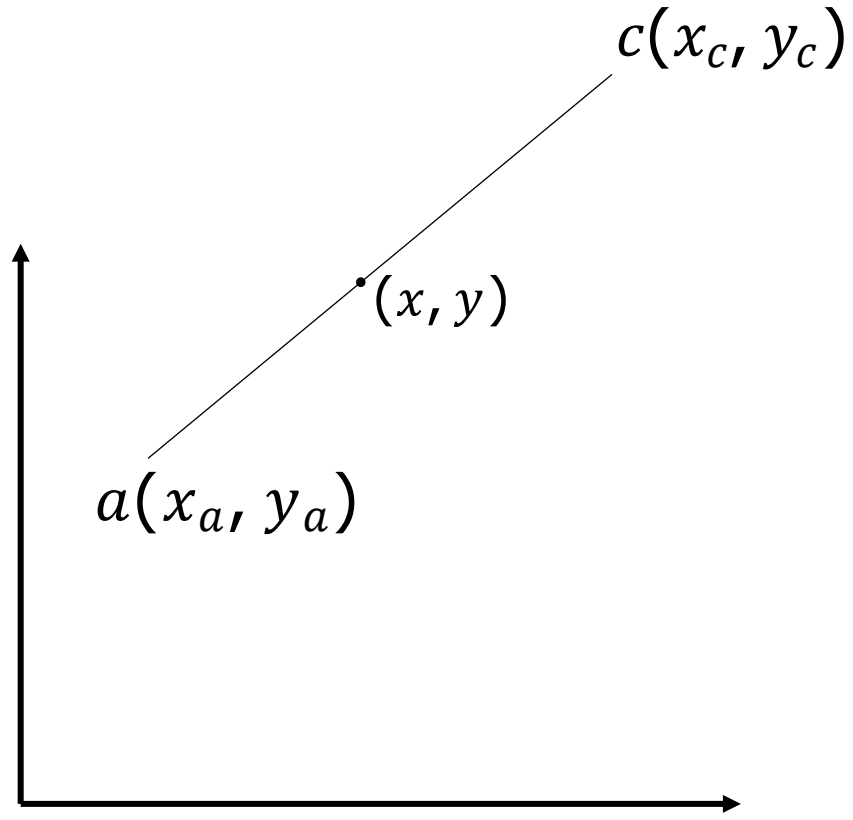
$$\Rightarrow y - \frac{y_c - y_a}{x_c - x_a}\,x - b = 0$$

$$\Rightarrow (x_c - x_a)y - (y_c - y_a)x - (x_c - x_a)b = 0$$

$$\Rightarrow (x_c - x_a)y - (y_c - y_a)x - C = 0$$

$$\Rightarrow (y_a - y_c)x + (x_c - x_a)y - C = 0$$

Cartesian → Barycentric

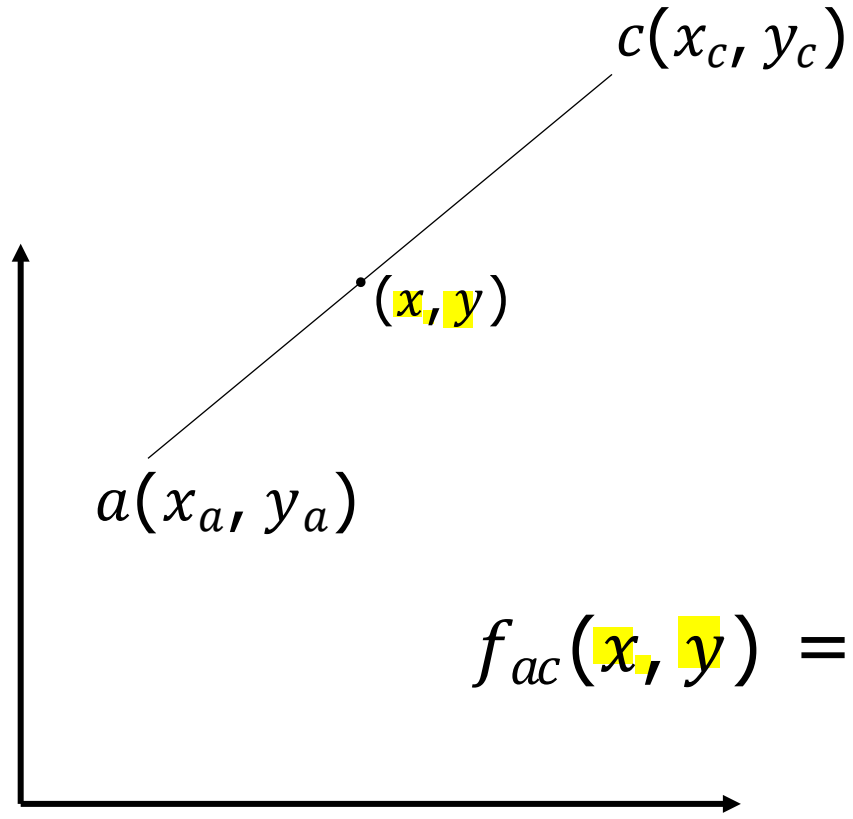


$$C = (y_a - y_c)x + (x_c - x_a)y$$

$$= (y_a - y_c)\boldsymbol{x_a} + (x_c - x_a)\boldsymbol{y_a}$$

$$= x_c y_a - x_a y_c$$

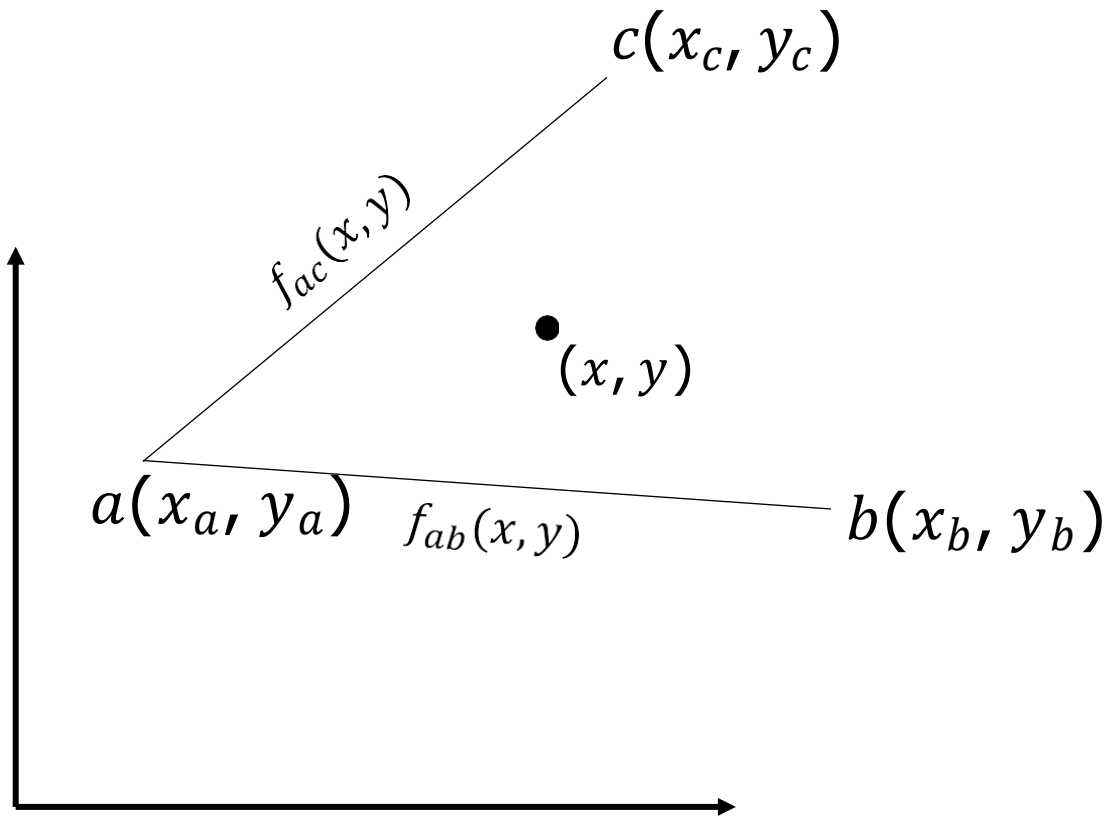
Cartesian → Barycentric



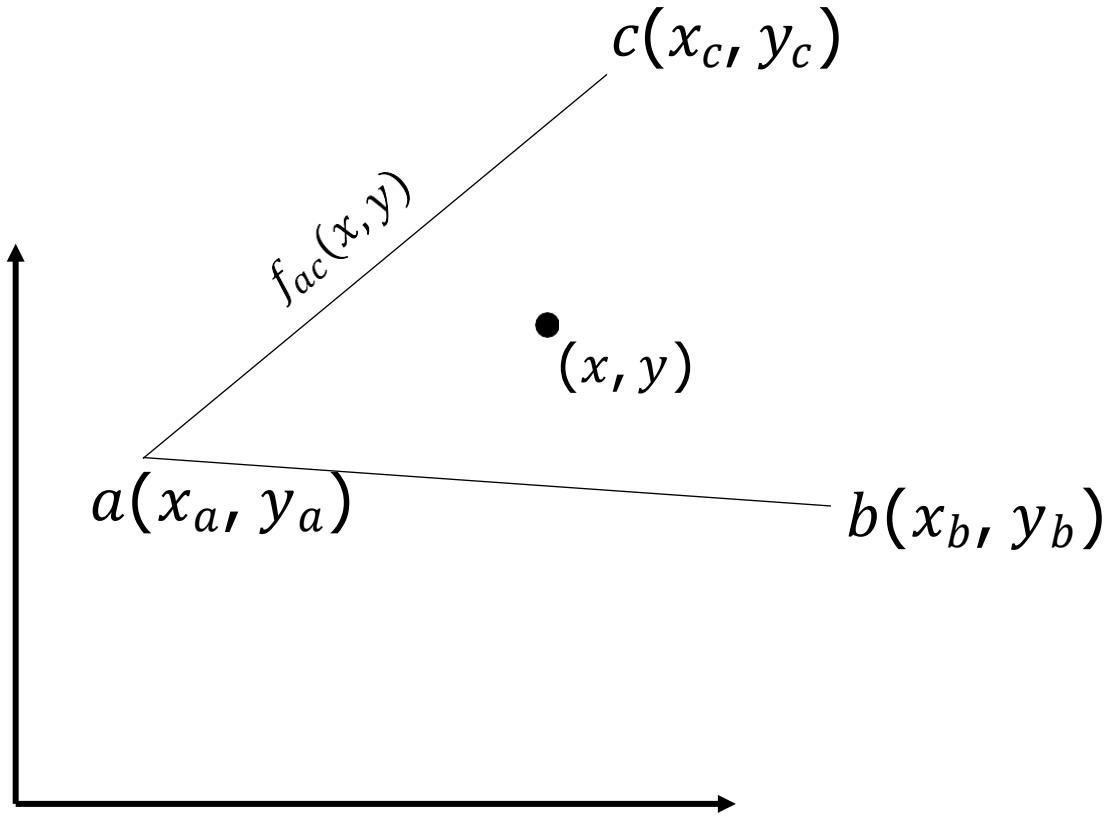
$$C = (y_a - y_c)x + (x_c - x_a)y$$
$$= (y_a - y_c)x_a + (x_c - x_a)y_a$$
$$= x_c y_a - x_a y_c$$

$$f_{ac}(x, y) = (y_a - y_c)x + (x_c - x_a)y - x_c y_a + x_a y_c = 0$$

Cartesian → Barycentric

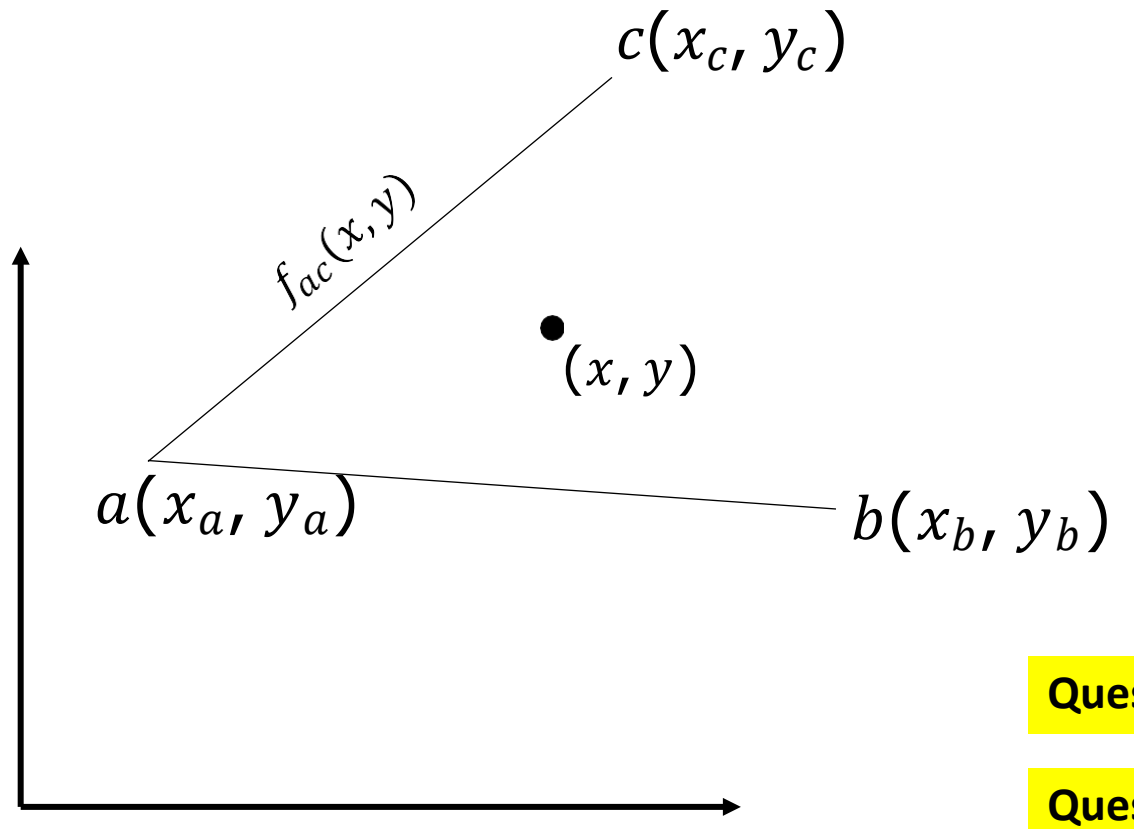


Cartesian → Barycentric



$$\beta = \frac{f_{ac}(x, y)}{f_{ac}(x_b, y_b)}$$

Cartesian → Barycentric



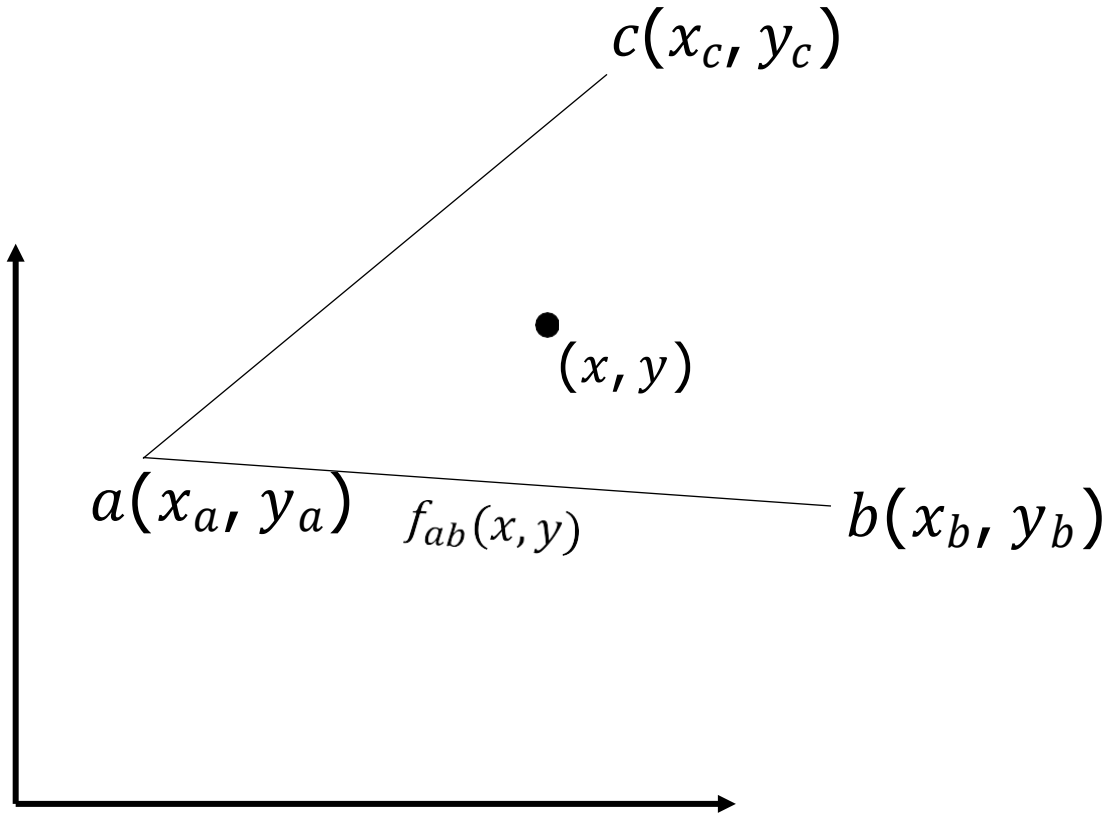
$$\beta = \frac{f_{ac}(x, y)}{f_{ac}(x_b, y_b)}$$

$$= \frac{(y_a - y_c)x + (x_c - x_a)y + x_a y_c - x_c y_a}{(y_a - y_c)x_b + (x_c - x_a)y_b + x_a y_c - x_c y_a},$$

Question – 1: In which case β becomes 1?

Question – 2: What will happen when *(x,y)* lies on $f_{ab}(x,y)$

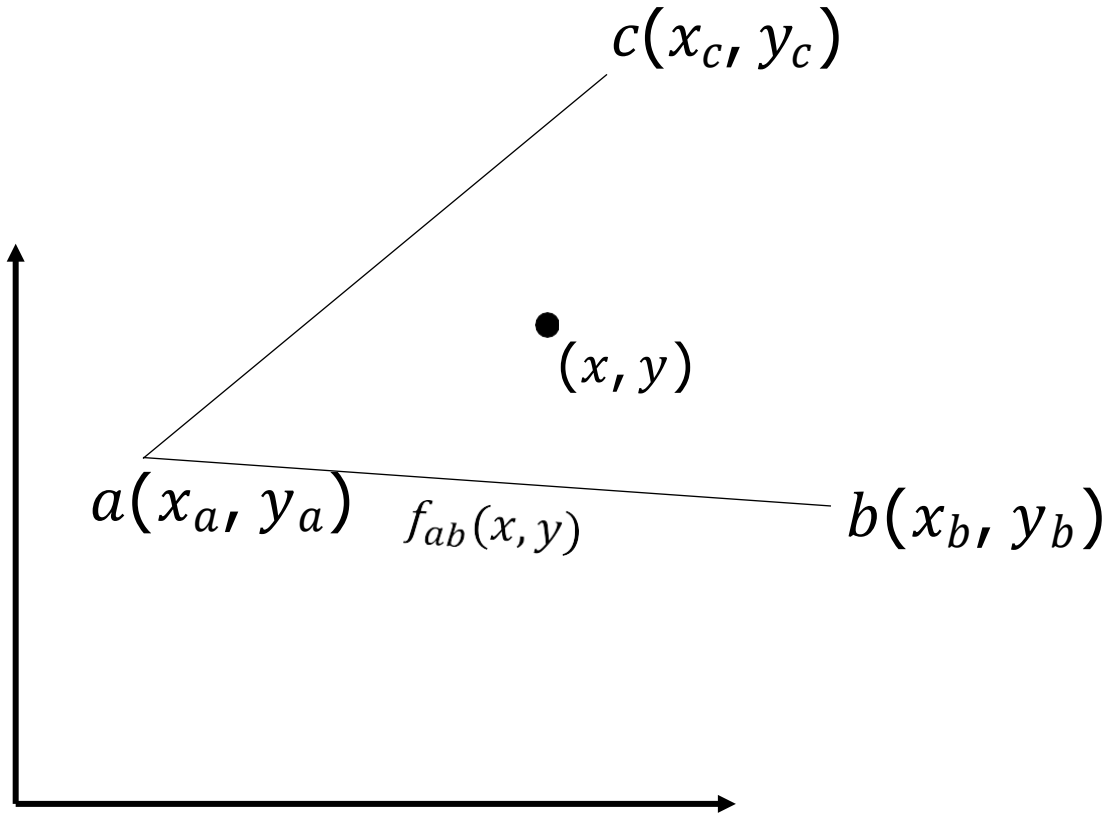
Cartesian → Barycentric



$$\beta = \frac{f_{ac}(x, y)}{f_{ac}(x_b, y_b)}$$

$$\gamma = \frac{f_{ab}(x, y)}{f_{ab}(x_c, y_c)}$$

Cartesian → Barycentric

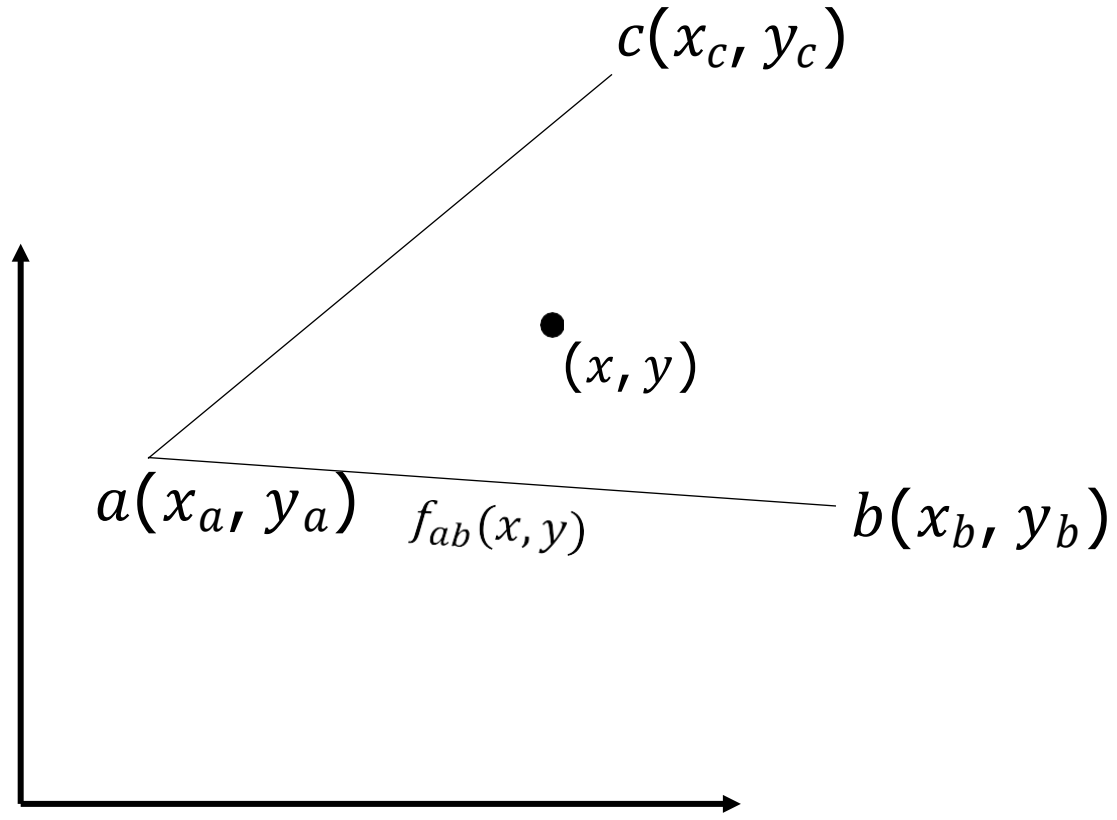


$$\beta = \frac{f_{ac}(x, y)}{f_{ac}(x_b, y_b)}$$

$$\gamma = \frac{f_{ab}(x, y)}{f_{ab}(x_c, y_c)}$$

$$\alpha = 1 - \beta - \gamma$$

Cartesian → Barycentric



$$P(x, y) \rightarrow P(\alpha, \beta, \gamma)$$

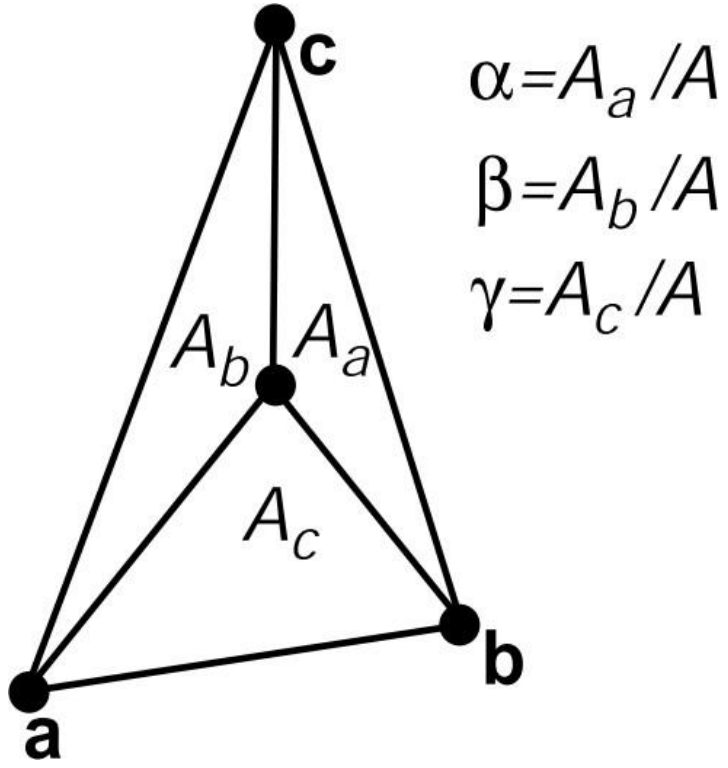
$$\beta = \frac{f_{ac}(x, y)}{f_{ac}(x_b, y_b)}$$

$$\gamma = \frac{f_{ab}(x, y)}{f_{ab}(x_c, y_c)}$$

$$\alpha = 1 - \beta - \gamma$$

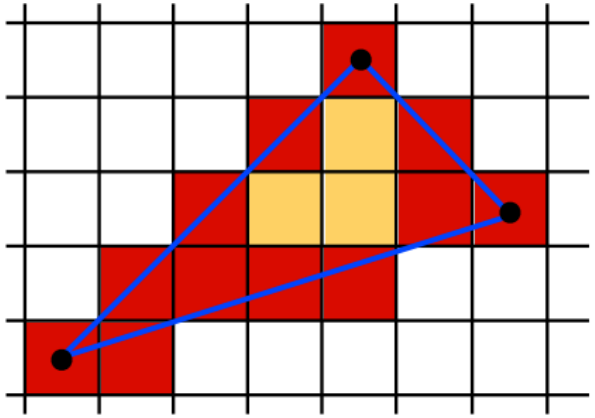
Barycentric Coordinate

Another approach:

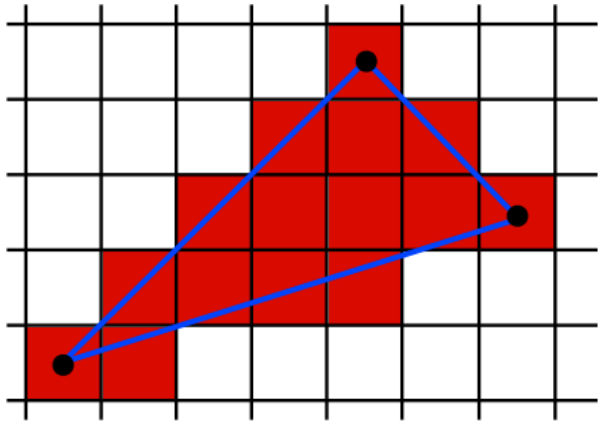


Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Triangle Rasterization (1/7)



Use Midpoint Algorithm for edges and fill in?



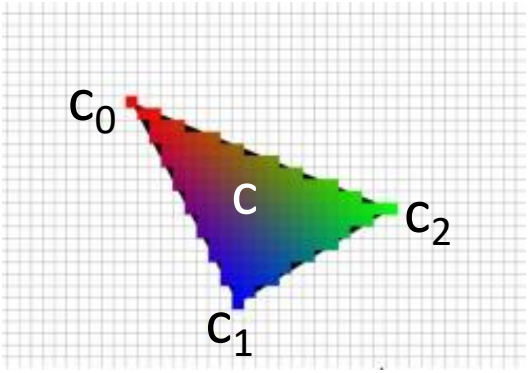
Use an approach based on barycentric coordinates

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Triangle Rasterization (2/7)

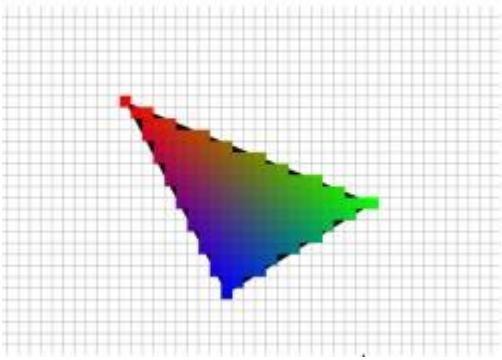
- If the vertices have colors c_0, c_1, and c_2, the color at a point in the triangle with *Barycentric coordinates* (α, β, γ) is:

$$\mathbf{c} = \alpha\mathbf{c}_0 + \beta\mathbf{c}_1 + \gamma\mathbf{c}_2$$



  - This type of interpolation of color is known in graphics as ***Gouraud interpolation***

Triangle Rasterization (3/7)



for all x **do**
  for all y **do**
    compute (α, β, γ) for (x, y)
    if $(\alpha \in [0,1]$ and $\beta \in [0,1]$ and $\gamma \in [0,1])$ **then**
      $\mathbf{c} = \alpha\mathbf{c}_0 + \beta\mathbf{c}_1 + \gamma\mathbf{c}_2$
      drawpixel (x, y) with color $\mathbf{c}$

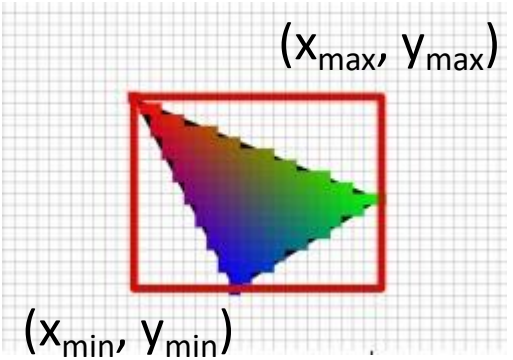
Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Triangle Rasterization (4/7)

for $y = y_{\min}$ to $y_{\max}$ **do**
  for $x = x_{\min}$ to $x_{\max}$ **do**

    compute (α, β, γ) for (x, y)

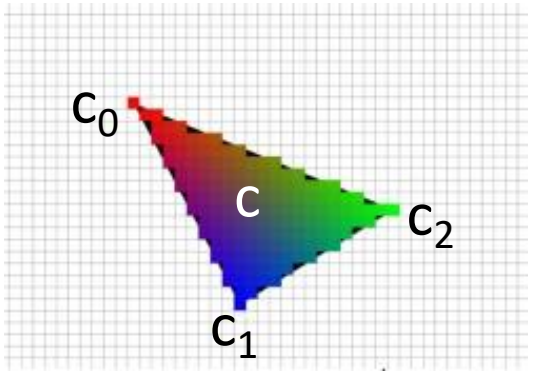
    if $(\alpha > 0$ and $\beta > 0$ and $\gamma > 0)$ **then**
      $\mathbf{c} = \alpha\mathbf{c}_0 + \beta\mathbf{c}_1 + \gamma\mathbf{c}_2$
      drawpixel (x, y) with color $\mathbf{c}$



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Triangle Rasterization (5/7)

for $y = y_{\min}$ to $y_{\max}$ **do**
  for $x = x_{\min}$ to $x_{\max}$ **do**
    $\alpha = f_{12}(x, y)/f_{12}(x_0, y_0)$
    $\beta = f_{20}(x, y)/f_{20}(x_1, y_1)$
    $\gamma = f_{01}(x, y)/f_{01}(x_2, y_2)$
    if $(\alpha > 0$ and $\beta > 0$ and $\gamma > 0)$ **then**
      $\mathbf{c} = \alpha\mathbf{c}_0 + \beta\mathbf{c}_1 + \gamma\mathbf{c}_2$
      drawpixel (x, y) with color $\mathbf{c}$

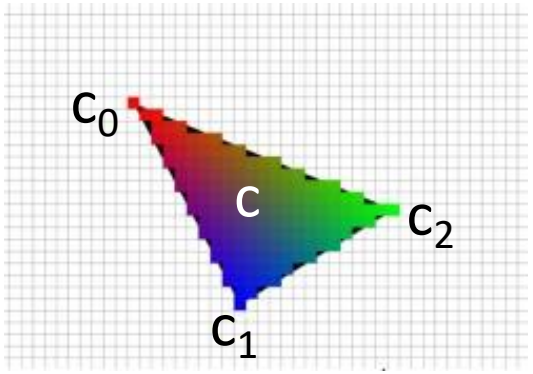


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Triangle Rasterization (6/7)

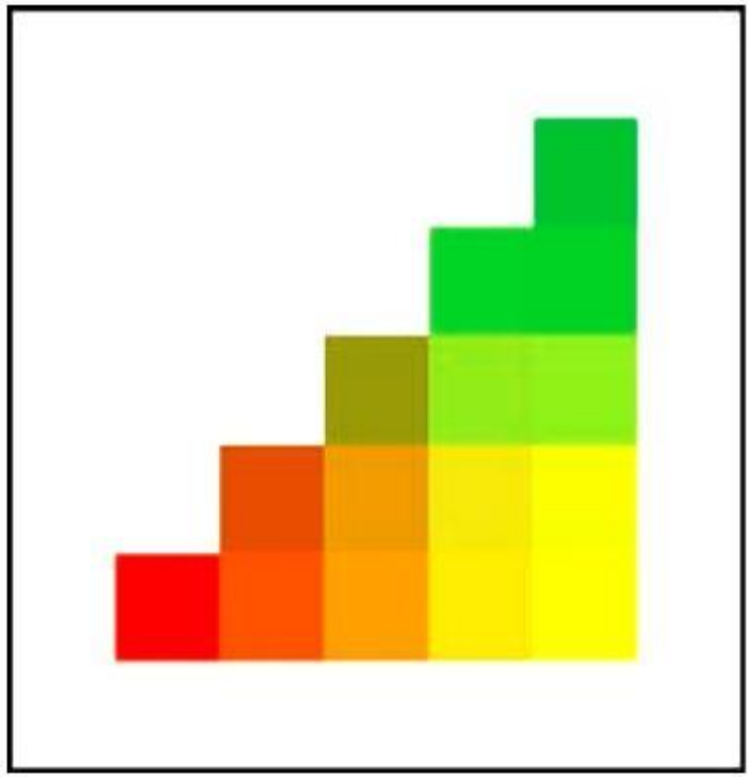
$$f_{01}(x,y) = (y_0 - y_1)x + (x_1 - x_0)y + x_0y_1 - x_1y_0,$$
$$f_{12}(x,y) = (y_1 - y_2)x + (x_2 - x_1)y + x_1y_2 - x_2y_1,$$
$$f_{20}(x,y) = (y_2 - y_0)x + (x_0 - x_2)y + x_2y_0 - x_0y_2.$$

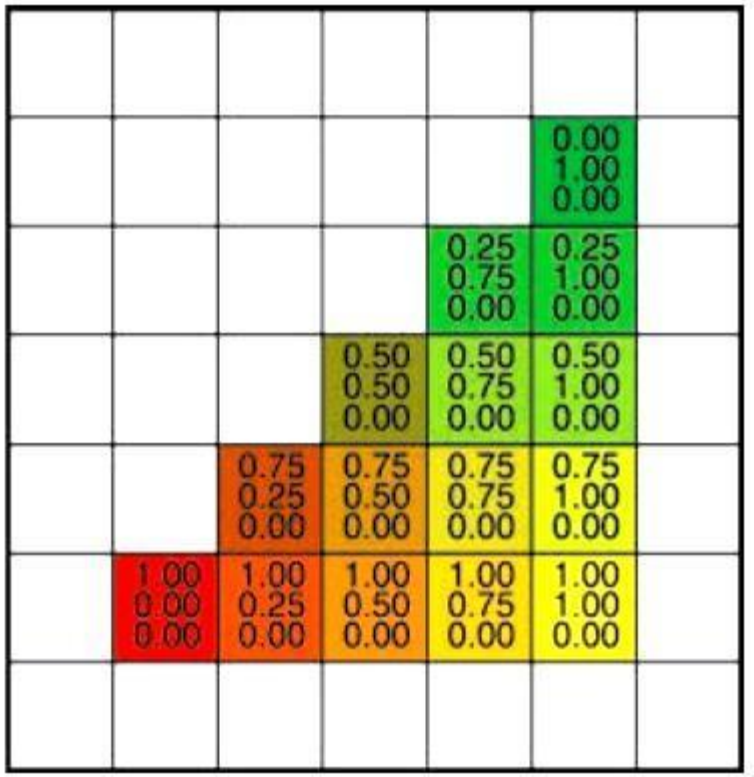
for $y = y_{\min}$ to $y_{\max}$ **do**
  for $x = x_{\min}$ to $x_{\max}$ **do**
    $\alpha = f_{12}(x,y)/f_{12}(x_0,y_0)$
    $\beta = f_{20}(x,y)/f_{20}(x_1,y_1)$
    $\gamma = f_{01}(x,y)/f_{01}(x_2,y_2)$
    if $(\alpha > 0$ and $\beta > 0$ and $\gamma > 0)$ **then**
      $\mathbf{c} = \alpha\mathbf{c}_0 + \beta\mathbf{c}_1 + \gamma\mathbf{c}_2$
      drawpixel (x,y) with color $\mathbf{c}$



Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Triangle Rasterization (7/7)





Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | http://www.cs.cornell.edu/courses/cs4620/2019fa/

Practice Problem

Consider a triangle with vertices A(2, 1), B(6, 1) and C(4, 5) and color values of red(0.5, 0, 0), green(0, 1, 0) and blue(0, 0, 1) at each vertex of the triangle. Find the color of the point P(3, 2) inside the triangle using the concept of Barycentric interpolation.

Further Reading

- Fundamentals of Computer Graphics, 4th Edition - Chapter 8